Actual quartic gluing on the fixed Jacobians

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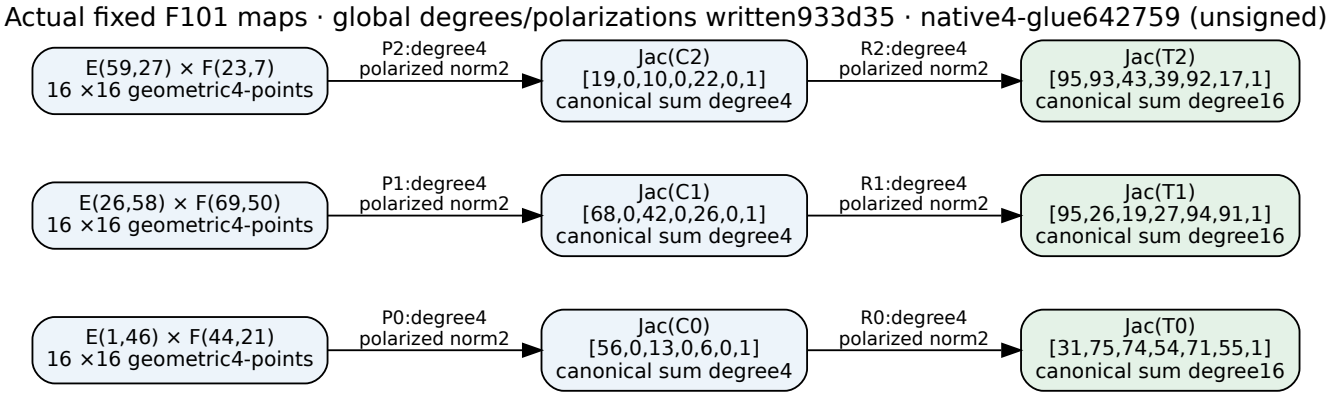
The new native census checks all256 elliptic-product4-torsion inputs on each T0,T1,T2. Each actual product pullback has kernel16 and image16, with bijective projections to both16-point elliptic factors. Every256 kernel-pair product has inverse factor Weil pairings. Complete changed standalone31 passes with zero failures,ignores or filters;old source guards,formatting,strict Clippy and all14 source/fixture hashes pass. PCC642759 retains the complete36,799-byte raw stream as standard XZ7,248, with both digests and exact roundtrip. [pcc:unverified]



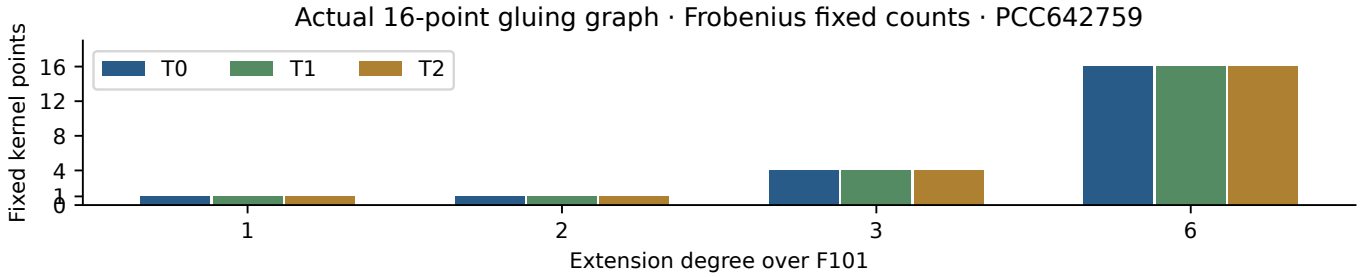
Ramified fibres are included

For nonzero elliptic2-torsion P=(X,0),the old ordinate inverse is unavailable. The new wrapper computes g=gcd(F,N−XQ²). In every case g has degree2 and (N−XQ²)/g is the square of a linear polynomial. The conjugate pair cancels in Picard,leaving the balanced class(g,0,0). If g contains the chosen branch W,the checker converts the remaining Weierstrass point directly to its point-minus-W quintic class. It keeps all three nonzero 2-points and exactly one chart-W corner per factor. [pcc:unverified]

All16 pulled classes per factor are valid and4-annihilated;each complete 256-entry addition table passes. Existing nonzero-ordinate pullback and chart bodies are unchanged. Field,polynomial,elliptic and separate Cantor engines are shared explicitly;producer Balanced/Richelot/trace code is unused. The written divisor calculation is in QUARTIC_GLUING_PROOF.md, PCC933d35. [pcc:unverified]

Pairings and negative controls

Order4 bases span all16 points. Miller’s two-doubling formula retains vertical factors;its normalized basis ratio matches the evaluation on shifted disjoint divisors. Each basis value is a primitive fourth root in F101. Bilinearity supplies the complete kernel-pair table. Inverting only the second factor rejects exactly96 of256 pairs per model. The initial128 prediction was a pre-native counting error;immutable protocolV1 and correctedV2/2c95c6 retain it explicitly. [pcc:unverified]



The actual order16 bound and six distinct surfaces

Admit the old actual P_i,R_i constructions,each of degree4 and polarized norm2,plus their certified factor embeddings. Then $K_i=R_iP_i$ has degree16, polarized norm4 and kernel in$(E_i\times F_i)[4]$. Its factor restrictions are embeddings,so both kernel projections are injective;equal order16 makes them bijective. Integral Tate-pairing functoriality makes the graph isotropic and hence an anti-isometry. The finite census independently checks the complete graph using explicit points. These are actual (4,4)-split principal Jacobians under the stated global dependencies. [pcc:unverified]

Sharpness on the actual Tate modules

An embedded connected elliptic subvariety has a torsionfree quotient Tate module,so its image is saturated. Its rational span is the corresponding Frobenius isotypic space. Thus the actual factor images equal M∩W109 and M∩W113. Milne’s isogeny Tate-cokernel formula gives index16 for their sum inside $M=T_2(\mathrm{Jac}(T_i))$,using $\deg K_i=16$. CT41′s injection bounds this quotient by$(Z/4)^2$;equal order makes it precisely$(Z/4)^2$. [pcc:unverified]

This realizes the external abstract lattice bound on each actual principally polarized Jac(T0),Jac(T1),Jac(T2). It uses the old global embeddings and degree,connected quotient exactness and the primary Tate theorem. It is not inferred from kernel counts alone. The common full4-torsion graph also attains Howe–Lauter’s gluing exponent4. [pcc:unverified]

All six geometric abelian varieties

The prior ordinary endomorphism-field argument b05286 gives cross-Hom0 and unique connected E-type/F-type elliptic subvarieties. Across distinct indices,the actual E j-values12,87,47 distinguish the surfaces. For the same index,the canonical sum of these intrinsic subvarieties has degree4 on $\mathrm{Jac}(C_i)$ and16 on $\mathrm{Jac}(T_i)$. Any surface isomorphism induces a bijection of those addition kernels,forcing equal degree. Therefore all15 pairs among the six unpolarized geometric Jacobians are nonisomorphic under these written geometric hypotheses. No native isomorphism solver or complete formal variety construction is asserted. [pcc:unverified]

Faithful formal and native algebra checks

Five bound localpass statements in QuarticGluing prove the universal duplication numerator,direct kernel-projection injectivity,kernel transport through a commuting square,quotient-kernel saturation and equality of saturated lattices with the same rational span. Claims d6878f/8b6f8c/ c7cd22/47fb74/0669e7;gate02 checks five with zero trusted native proofs. The written census is154 declarations across31 modules. Actual Picard, polarization,Tate exactness and classification remain written bridges. [pcc:unverified]

The unchanged repository identity/SHA checker issues IDC1he156b6adc8f9cdd5 with32 evaluations and64 extra domain-separated checks over$2^{61}-1$. It pins the exact Lean source. All23 algebra tests pass, including a rejected identity mutation;1176bb preserves raw/source/hash evidence. Finite evaluation is separate from the universal Lean proof. The new CI job checks both imported-source digests and exact certificate. [pcc:unverified]

Sources,failures and next step

Primary Howe–Lauter §2/MilneI7 andIV2.2(f07968 plus Tate review),MilneI13.1–2 and Galbraith26.2/26.3.1(8b9b20) were read. External CT41 atca083268 is credited. KB/wrapper and Conductor are unavailable;local accepted proofs and primary papers were used. No novelty claim. [pcc:unverified]

Static9bfde8,Lean/API+dependency02a50b,legacy styleb93d1e and the initial missing-PATH binding receipt remain. Failed builds are never counted. Prior reports retain their historical scopes;these new figures depict actual4-gluing and the six-surface invariant. Next:transport this graph along the nondual7-chain,then test bounded elliptic-cycle closure. Current head needs its own exact CI;benchmark admission still lacks a live balance and the atlas vault is locked. [pcc:unverified]